温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3001 WHS01
Python Programming for Business

Introduction to Python

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Week 2-1, 3 March 2026

Key Revisions to the Syllabus

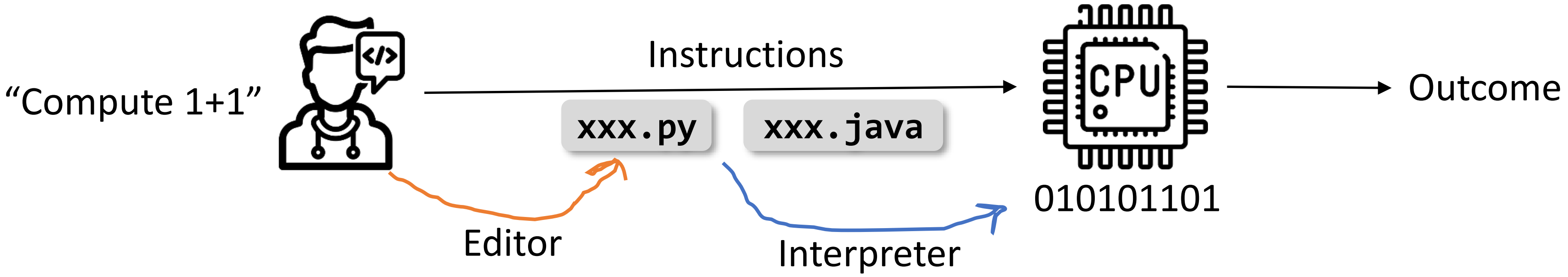
- Final Project (30%)
  - 15% **Paper**: Conference-style empirical study (15–20 pages including everything, 12-point font, double-spaced), due on 6/09 at 23:59
  - 15% **Presentation:** ~10mins presentation during the last week (6/04 & 6/09)
- Midterm Exam (30%)
  - Closed-book, in-class written exam (code interpretation + conceptual understanding)
  - Tentative date: **4/14** (at least one week before the early alert DDL)
- Project Milestones – 3 Assignments (30%)
  - Assignment 1 (**3/15**): Proposal with research question, motivation, research gap
  - Assignment 2 (**4/05**): Revised proposal with hypotheses, data and empirical strategy
  - Assignment 3 (**4/30**): Data collection report and dataset submission

Outline

- What is Programing?
- Why Python?
- Python Installation
- From Console to IDE
  - Install VS Code
  - Install Python and Jupyter extension in VS Code
  - First Python script (.py), Jupyter notebook (.ipynb)
- English vs. Programming Language
  - Primitive constructs, syntax, static semantics, semantics

What Is Programming?

- **Programming** is the process of giving precise instructions to a computer to perform tasks.
- Why do we need programming?
- A **programming language** is the formal system of keywords and rules we use to write instructions for a computer.
- A **program** is a sequence of:
  - Definitions (e.g., variables, functions)
  - Commands (statements) that instruct the computer to perform actions



Why Python?

- Different Languages, Different Purposes & Strengths
  - **JavaScript** → Web development
  - **Python** → Data analysis, AI, automation
  - **C/C++** → System programming, embedded systems
  - …

- Python
  - High-level programming language
  - **Simple**, **readable syntax**
  - **Rich ecosystem** for data and AI (“包/库”很多!)
  - Widely adopted in finance, consulting, technology
  - Beginner-friendly, yet scalale for advanced analytics

Python Installation

How do we save, organize, and rerun code in a professional workflow?

- Step 1: Download Python
  - Go to: https://python.org
  - Download the version that matches your operating system (Windows / macOS)
- Step 2: Install
  - Run the installer
  - Check "**Add Python to PATH**" (Important)
  - Click **Install Now**
- Step 3: Verify Installation (Command Line)
  - Open Command Prompt (Windows) or Terminal (macOS)

🎉 Interact with Python in command line

```
python --version
python3 --version
python
exit()
where python (windows)
which python (macOS)
```

From Console to IDE

[VS Code Documentation]

[Learn Visual Studio Code in 15 minutes: 2026 Official Beginner Tutorial]

- Integrated Development Environment (IDE)
  - A tool that helps you write code, run programs, debug errors, manage files efficiently
  - *Pros:* better readability, error detection, auto-completion, file organization, professional workflow
  - *Examples:* VS Code, Pycharm, Jupyter Notebook

- Setting Up VS Code
  - Step 1: Download and install VS Code from https://code.visualstudio.com/
  - Step 2: Install Python extension
    - Inside VS Code: Go to Extensions → Search for Python → Install the official Python extension
  - Step 3: Your first Python script (`.py`)
    - Create a new file → `hello.py`
    - Write: `print(“Hello, Python!”)`
    - Save the file to your working directory
    - Run the script
    - Observe the output in the terminal window

Environment Setup

[Project Jupyter Documentation]

- Setting Up VS Code
  - Step 4: Install Jupyter extension
    - Inside VS Code: Go to Extensions → Search for Jupyter → Install the extension
  - Step 5: Your first Jupyter notebook (`.ipynb`)
    - Create a new file → `analysis.ipynb`
    - Add a code cell
    - Write: `print(“Data Analytics Starts Here.”)`
    - Save the file to your working directory
    - Run the cell
    - Observe the output

How does this differ from executing a Python script (`.py`)?

Jupyter Notebook allows:
- Code + explanation
- Step-by-step execution
- Visualization integration

English vs. Programming Language (PL)

	English	PL
Primitive Constructs (Basic Building Blocks)	• Words • Example: *`cat, dog, boy`*	• Numbers: `10, 3.14` • Strings: “`hello`” • Operators: `+, -, *, /`

Just as sentences are built from words, programs are built from primitive elements.

English vs. Programming Language (PL)

		English	PL
Syntax	Rules for well-formed structure	• *“cat boy hugs”* • *“cat hugs boy”*	• `“hi” 5` • `“hi” * 5`
Static Semantics	Rules for type & consistency correctness before execution	`“The table eats pizza.”`	`“hi” + 5`
Semantics	The actual meaning of the sentence or program	`“The chicken is ready to eat.”`	A program has one precise meaning, but it may not be the meaning the programmer intended.

Summary

- Programming is giving precise, logical instructions to a computer.
- Python is widely used in data analytics, AI, and automation.
- We installed Python and verified it via the console.
- We introduced VS Code as a professional development environment.
- We created and executed:
    - A Python script (`.py`)
    - A Jupyter notebook (`.ipynb`)
- Programming languages are built from primitive constructs.
- Code must follow strict syntax rules.
- A program has one precise meaning — but it may differ from the programmer intention.